

A Local–Geometry Changepoint Aligned with the Onset of Portuguese Grammatical Competence in a Small Language Model

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Abstract

Capabilities can emerge abruptly during training and be missed by smooth loss curves and behavioral evaluations — a core AI-safety concern. We ask whether a structural signal from singular learning theory (SLT), the *local learning coefficient* (LLC, $\hat{\lambda}$, estimated by SGLD), tracks capability emergence in a small, fully controlled setting. Adapting **TinyStories-8M** to Portuguese by continued pretraining, we measure at every checkpoint both behavior (bits-per-byte; a 538-item grammar benchmark) and the LLC. Modeling loss falls smoothly, but grammatical competence emerges *abruptly* (chance $\rightarrow$ 89%, the preference margin flipping at $\approx$ 2.5M tokens); over exactly this window the LLC rises steeply ($+52 \rightarrow +87$) then plateaus, slightly *leading* behavior. Two controls (shuffled-Portuguese, matched-English) do not reproduce this, establishing specificity, and the shape is robust to the SGLD localization. We frame this cautiously as a changepoint in an SLT-derived geometric estimate aligned with a behavioral transition — a proof-of-concept for behavior-independent, training-time signals of capability acquisition.

New work statement. Every result, figure, control, and the supporting code in this report were produced during the hackathon (2026-06-19 to 06-21). We build on existing SLT/developmental-interpretability ideas and the open-source **devinterp** LLC estimator (Section 2); the experimental design, the Portuguese-acquisition study, the controls, and all measurements are new.

1 Introduction

Why this matters for AI safety. A central difficulty in monitoring advanced models is that capabilities can appear *suddenly* during training or scaling, and a smooth training-loss curve gives little warning. Behavioral evaluations are our main detector, but they are reactive — they can only find a capability once it is expressible and once someone thinks to elicit it, and they can miss latent or sandbagged abilities. A complementary goal is a *behavior-independent, training-time* signal that an internal capability is forming: a structural “early-warning” read on the model itself. Developmental interpretability pursues exactly this — tracking internal, structural quantities across training to see whether they change when capabilities do.

Singular learning theory (SLT) provides one such quantity: the *local learning coefficient* (LLC, also called the local RLCT), a measure of the effective dimensionality / degeneracy of the loss

landscape around a parameter [1, 2]. If the geometry reorganizes precisely as a capability appears, the LLC should exhibit a changepoint aligned with — or even ahead of — the behavioral one.

We study this in a deliberately small, controllable setting: second-language acquisition by continued pretraining. We take an English-trained model (**TinyStories-8M**) and adapt it to Portuguese, measuring behavior and the LLC along the whole trajectory. **The model learning Portuguese is not the result — it is the scaffolding.** The contribution is the *LLC $\leftrightarrow$ behavior alignment*, established together with the controls that make it specific. Our findings:

- Behavior shows a clean grammatical transition at $\approx 1.5\text{--}2.5\text{M}$ tokens even though bits-per-byte falls smoothly (Section 4.1).
- The LLC rises steeply through that window and then plateaus — a changepoint aligned with the behavioral one (Section 4.2).
- Two controls (shuffled-PT, matched-English) fail to reproduce the signature, so it is specific to genuine language structure being acquired, not to training length or generic adaptation (Section 4.3).
- The trajectory shape is robust to the SGLD localization hyperparameter (Section 4.4).

Theory of change. If a structural quantity reliably exhibits a changepoint when — and slightly before — a capability emerges, and if that signal is specific (controls do not trigger it), then SLT-derived measurements could become a component of *developmental monitoring*: watching training for the formation of capabilities before they are behaviorally elicited, complementing behavioral evals. This report is a controlled proof-of-concept of the core link (an LLC changepoint aligned with, and leading, a behavioral one), not a deployable monitor; we are explicit about that gap in Section 6. We also note a practical angle relevant to a Global-South research context: the entire study runs on a single small GPU at trivial cost (an 8M-parameter model, $\sim \$15$ of compute), so this line of work is accessible without frontier-scale resources.

2 Related Work

Singular learning theory and the LLC. SLT characterizes the loss landscape of singular (non-identifiable) models such as neural networks [1]; the (local) learning coefficient quantifies local degeneracy. Recent work makes the LLC estimable at scale via SGLD [2], which we use through the open-source **devinterp** library. **Developmental interpretability.** The closest line tracks the LLC and related quantities across training to identify developmental stages and structure formation — e.g. the developmental landscape of in-context learning [3] and dynamical/Bayesian transitions in toy models [4]. **What is new here.** Prior LLC-during-training studies focus on in-context learning or toy/algorithmic tasks; we instead study *second-language acquisition* by continued pretraining, pair the LLC trajectory with a dense behavioral grammar-acquisition curve, and — crucially — add *content-matched controls* (shuffled-Portuguese, matched-English) and a localization-robustness check that together establish the changepoint is specific to genuine structure acquisition. To our knowledge this specific alignment-with-controls in a language-acquisition setting is new.

3 Methodology

Model and training. We adapt **roneneldan/TinyStories-8M** by full-parameter, FP32 next-token continued pretraining (AdamW, learning rate 10^{-4} , 3% warmup, cosine decay, gradient clipping

1.0, weight decay 0.01, batch size 64, sequence length 128) on a Portuguese Wikipedia corpus, to 100M tokens, with 17 log-spaced checkpoints. The recipe was fixed by a short “autoresearch” search after an initial under-trained run; we then froze it across all conditions.

Conditions. Following the minimal design for this question we run (i) *structured Portuguese* (the condition of interest), and two controls: (ii) *token-shuffled Portuguese* — the same tokens with within-sequence order destroyed, so there is no grammar to acquire — and (iii) *matched English* — the identical protocol on English, controlling for “any continued pretraining raises complexity.”

Behavioral metrics (every checkpoint). Portuguese validation bits-per-byte (BPB), English-retention BPB, and a frozen, hashed **538-item templated Portuguese grammatical minimal-pair benchmark** (10 agreement phenomena). Accuracy is the fraction of pairs for which the model assigns higher log-probability to the grammatical sentence; we also report the mean log-prob margin.

LLC estimation. We estimate the LLC by SGLD [2, 3] using the `devinterp` library, with one globally fixed sampler configuration across each trajectory (3 chains, learning rate 10^{-5} , $n_\beta=10$, localization 100, batch 64, 200 burn-in + 100 draws). With L the loss and L_0 the loss at the checkpoint center,

$$\hat{\lambda} = n_\beta (\mathbb{E}_{\text{SGLD}}[L] - L_0), \quad (1)$$

which is positive at a true local minimum of the measured loss. **Crucially, each condition is measured against a reference set drawn from *its own* training distribution** (full-length, non-padded chunks): $\hat{\lambda}$ is only valid at a minimum of the loss the model actually minimized, so using one condition’s data to score another would yield meaningless estimates.

A measurement note (why the controls matter twice). An early version of our LLC pipeline built its reference from short sentences padded to the sequence length without masking, so $\approx 90\%$ of every scored sequence was unmasked padding. This produced *negative* $\hat{\lambda}$ everywhere — the checkpoint is not a minimum of *that* (padded) loss. The fix was to score non-padded chunks of the actual training distribution. We flag this because it is the central methodological lesson: an LLC estimate is only meaningful relative to a loss the model truly minimized, and one must read the *value, sign and chain agreement*, never a binary “accepted” label.

4 Results

4.1 Behavior: smooth loss, abrupt grammar

The model genuinely acquires Portuguese: PT BPB falls monotonically $6.69 \rightarrow 2.54$ with no kink (Figure 1). Grammatical competence, by contrast, emerges abruptly: minimal-pair accuracy sits at chance through 1.5M tokens, the mean margin flips negative→positive between 1.5M and 2.5M, and accuracy then climbs to 89.2% by 27M (Table 1). The abruptness lives in the discrete grammatical behavior, not in the modeling-loss curve.

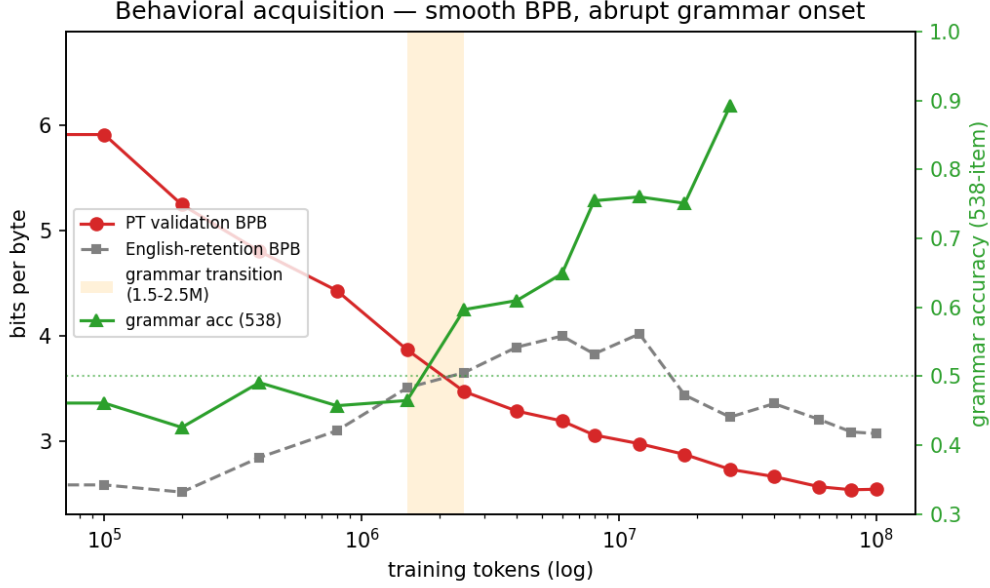


Figure 1: Behavioral acquisition (structured PT, seed A). PT BPB (red) falls smoothly; the 538-item grammar accuracy (green, right axis) is flat at chance then rises sharply through the shaded 1.5–2.5M window. English-retention BPB (grey) degrades then partly recovers.

tokens	PT BPB	EN BPB	grammar acc.	grammar margin
0	6.686	2.714	0.446	−0.277
1.5M	3.869	3.506	0.465	−0.394
2.5M	3.471	3.651	0.597	+0.461
8M	3.057	3.829	0.755	+1.540
27M	2.731	3.228	0.892	+2.403
100M	2.541	3.071	—	—

Table 1: Behavioral trajectory (structured PT, seed A). The grammar margin crosses zero at ≈ 2.5 M tokens. Grammar was scored to 27M (the transition is well before); BPB to 100M.

4.2 The LLC trajectory: a rise-then-plateau changepoint

The LLC is positive at every checkpoint, with tight chain agreement and no rejected chains. It rises ≈ 34 points over 400k $\rightarrow$ 4M (steepest through 400k–2.5M) and is then flat within noise (≈ 84 –87) for the remaining $25\times$ of training (Figure 2, left; Table 2). The steep rise *brackets* the behavioral grammar transition: the geometry reorganizes through ≈ 400 k–2.5M, the grammar margin crosses zero at ≈ 2.5 M, and both then settle (Figure 2, right). Read carefully, the LLC rise slightly *leads* the behavioral expression — local complexity increases first, grammatical behavior follows and continues maturing after $\hat{\lambda}$ has plateaued.

tokens	400k	800k	1.5M	2.5M	8M	100M
$\hat{\lambda}$ (seed A)	+52.3	+61.5	+75.5	+80.7	+86.0	+86.7

Table 2: Primary LLC trajectory (structured PT, seed A; loc 100, 3 chains). Full 11-checkpoint trajectory in Figure 2.

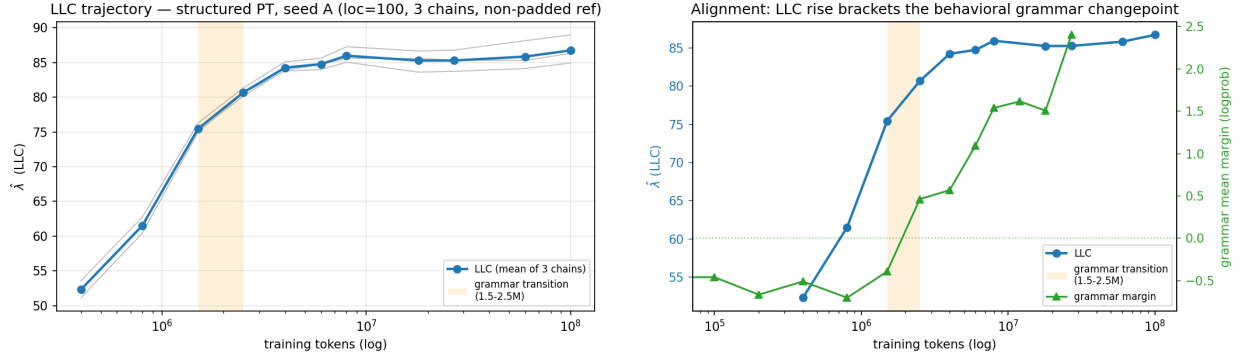


Figure 2: Left: the LLC trajectory (structured PT, seed A) rises steeply then plateaus; thin grey lines are individual chains. **Right:** alignment — the LLC rise (blue) brackets the grammar-margin sign change (green) in the shaded 1.5–2.5M window.

4.3 Controls: the changepoint is specific

Neither control reproduces the rise-then-plateau (Figure 3, Table 3). The token-shuffled model rises only weakly to ≈ 60 and then *declines* to ≈ 28 as it settles into low-complexity position-independent statistics. The matched-English model genuinely adapts (complexity rises monotonically $51 \rightarrow 70$) but *smoothly and to a markedly lower level*, with no steep transition and no high plateau — the key control showing the signature is not merely “continued pretraining raises the LLC.” At 100M the three conditions are cleanly separated, and only structured Portuguese has both the steep grammar-window rise and the high plateau. The geometric changepoint is therefore specific to genuine Portuguese structure being acquired, not to training length, token statistics, or generic adaptation.

tokens	structured PT	shuffled PT	matched EN
1.5M	+75.5	+60.4 (peak)	+58.8
8M	+86.0	+43.2	+64.8
100M	+86.7	+27.9	+70.3

Table 3: LLC at shared checkpoints across the three conditions (loc 100, 3 chains, condition-matched references).

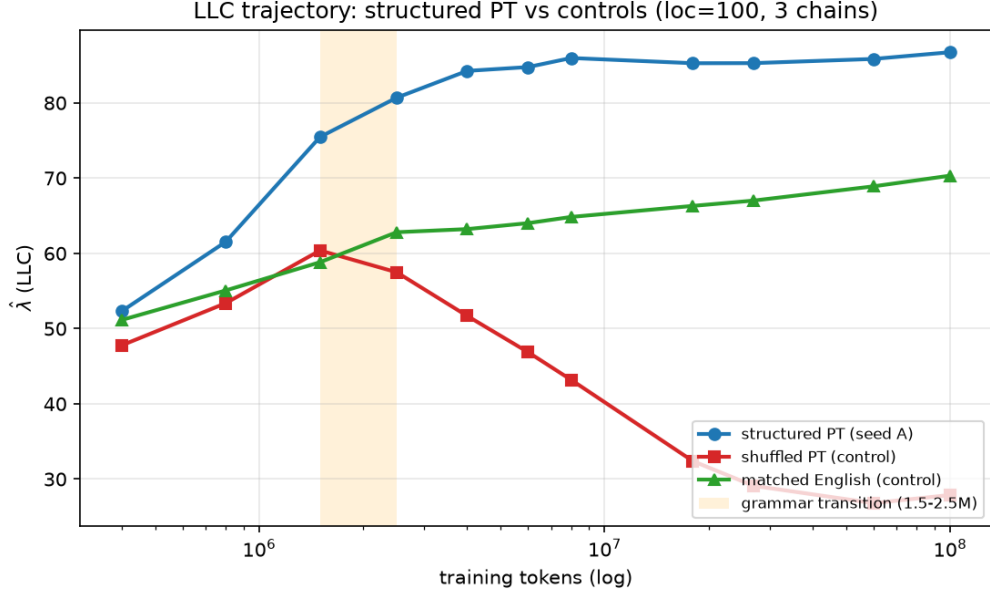


Figure 3: Three-condition contrast. Structured PT (blue) rises to a high plateau; shuffled PT (red) rises then declines; matched English (green) rises smoothly to a lower level. The shaded band marks the behavioral grammar transition.

4.4 Robustness to the localization hyperparameter

The SGLD localization γ sets the prior strength around the checkpoint, so it shifts the absolute level of $\hat{\lambda}$. At $\gamma = 300$ the structured-PT trajectory sits ≈ 4 units below $\gamma = 100$ everywhere but is *identical in shape* — same steep rise through 1.5–2.5M, same plateau (Figure 4). The changepoint is thus a property of the trajectory, not of the localization choice.

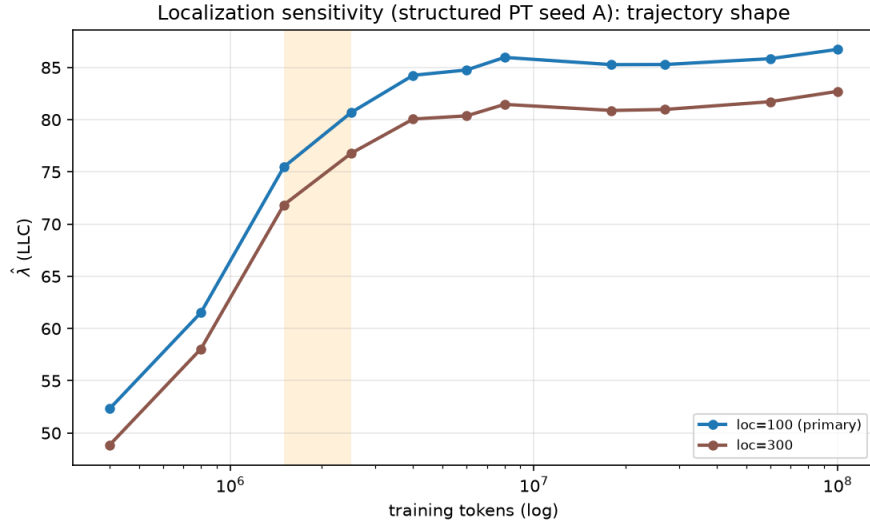


Figure 4: Localization sensitivity (structured PT, seed A): $\gamma = 100$ vs. $\gamma = 300$ are near-parallel — the rise-then-plateau shape is preserved.

5 Discussion

Across an identical architecture, optimizer, and token budget, three conditions produce three qualitatively different LLC trajectories, and only the one with real grammatical structure to acquire shows a steep rise bracketing the moment grammatical behavior appears. A geometric changepoint and a behavioral changepoint coincide where the modeling-loss curve shows nothing. Following standard caution we call this *a changepoint in an SLT-derived local geometric estimate aligned with a behavioral transition*, and we do not claim a formal SLT phase transition, nor grokking (training and held-out performance improve together). The slight *lead* of the LLC over behavior is suggestive of the geometry reorganizing before the capability is fully expressed, but we do not over-read a single trajectory.

6 Limitations and Dual-Use Considerations

Limitations.

- **Single Portuguese seed.** The controls establish specificity but vary data *content*, not the random seed; they do not rule out that the exact trajectory is idiosyncratic to one initialization / data order. A genuine second-seed replication (independent data ordering) is *in progress* and will be folded in.
- **Scale and transfer.** This is one 8M-parameter model on one language pair. Whether the alignment holds — and whether the LLC remains estimable and informative — at frontier scale, where the safety stakes lie, is untested and is the central open question for the monitoring application.
- **No formal changepoint test.** “The rise brackets the transition” and “the lead” are read from the trajectories, not from a fitted changepoint model with a significance statement.
- **LLC estimation is fragile and costly.** It is hyperparameter- and reference-sensitive (Section 3), has no ground truth, and each trajectory is an MCMC procedure; a deployable monitor would need cheaper, more robust estimation.
- **Localization sweep is two points** ($\gamma \in \{100, 300\}$) and the 538-item grammar set is template-generated, not a validated broad-coverage suite.

Dual-use considerations. The method is a measurement / monitoring tool and is primarily *defensive*: its intended use is to help detect capability formation, which supports oversight rather than capability creation. The misuse surface is limited but not zero: a reliable training-time signal that a capability is “forming” could in principle be used to guide or accelerate the development of a targeted (including undesirable) capability, or to probe for capabilities an actor wishes to elicit. We judge this risk low for a fragile, compute-heavy measurement on tiny models, and outweighed by the defensive value of structural early-warning signals. A more immediate risk is *epistemic*: a plausible-looking but mis-estimated LLC (e.g. the unmasked-padding bug we hit, or reading an “accepted” label instead of the value) could create false confidence; we therefore stress reading the value, sign, and chain agreement, and validating against controls. Nothing here lowers a misuse barrier at the frontier.

Future improvements. (i) Finish the second-seed replication and add more seeds; (ii) a formal statistical changepoint/alignment test; (iii) more capabilities and language pairs, and a wider

localization sweep; (iv) cheaper, more robust LLC estimation; (v) the key step for the safety case — testing whether the alignment holds at larger scale.

7 Conclusion

In a small, fully controlled language-acquisition setting, an SLT-derived local geometric quantity exhibits a changepoint aligned with the behavioral onset of grammatical competence, shown specific by two controls and robust to the SGLD localization. Pending a second-seed replication, this is, to our knowledge, a clean demonstration of an LLC $\leftrightarrow$ behavior alignment during second-language acquisition.

References

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